

A REVIEW ON TREND ANALYSIS AND HOTSPOT MAPPING METHODS ON COVID-19 (2019-2022) USING GEOGRAPHIC INFORMATION SYSTEMS

by

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AUTHOR'S DECLARATION

I, (Vaidya Sai Teja), declare that the research work carried out for this special study report was in accordance with the regulations of the Asian Institute of Technology. The work presented in it are my own and has been generated by me as the result of my own original research, and if external sources were used, such sources have been cited. It is original and has not been submitted to any other institution to obtain another degree or qualification. This is a true copy of the special study report, including final revisions.

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I would like to express my gratitude to AIT- Fellowship Scholarship for giving financial support to my special study in Remote Sensing and Geographic Information System in Asian Institute of Technology.

ABSTRACT

This study presents a comprehensive Geographic Information Systems (GIS)-based analysis of COVID-19 hotspots and trends. It highlights the need for a global perspective to understand the pandemic's multifaceted impact on healthcare systems, economies, and the environment. Beyond traditional epidemiological considerations, the study explores the pandemic's social, economic, and environmental dimensions, emphasizing that epidemics such as COVID-19 transcend geographical boundaries.

Chapter 2 reviews the applications of trend analysis methods in spatial epidemiology, discussing methodologies such as ARIMA models, Successive Approximation, Google Trend Analysis, and Microsoft Software to predict the number of COVID-19 cases and explore pandemic trends. The chapter also highlights the significance of data visualization tools in understanding and managing epidemics.

Chapter 3 focuses on hotspot analysis in the context of COVID-19 research, exploring spatial statistical techniques such as Spatial Correlation (Getis-Ord G_i^* Statistic), Machine Learning Methods (Gaussian Mixture Model), Kernel Density Estimation, and the use of open-source data. It demonstrates how hotspot analysis is used to identify areas with high concentrations of COVID-19 cases and how it can aid targeted resource allocation and containment strategies.

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CHAPTER 1

INTRODUCTION

This chapter represents the background and motive in performing research on the Hotspot mapping and trend analysis of Covid-19 influence all over the world (Asia, Africa, North America, South America, Antarctica, Europe, and Australia.). The first section provides the background knowledge about Covid-19 and the current situation of Covid-19 on Multi-countries. Problem statement along with research questions are presented in later sections.

1.1. Background

On December 31, 2019, the World Health Organization (WHO) was alerted to a cluster of pneumonia cases of unknown origin in Wuhan, China. The causative agent was identified as a novel coronavirus, SARS-CoV-2, on January 7, 2020. The disease caused by SARS-CoV-2 was named COVID-19 by the WHO on January 12, 2020.

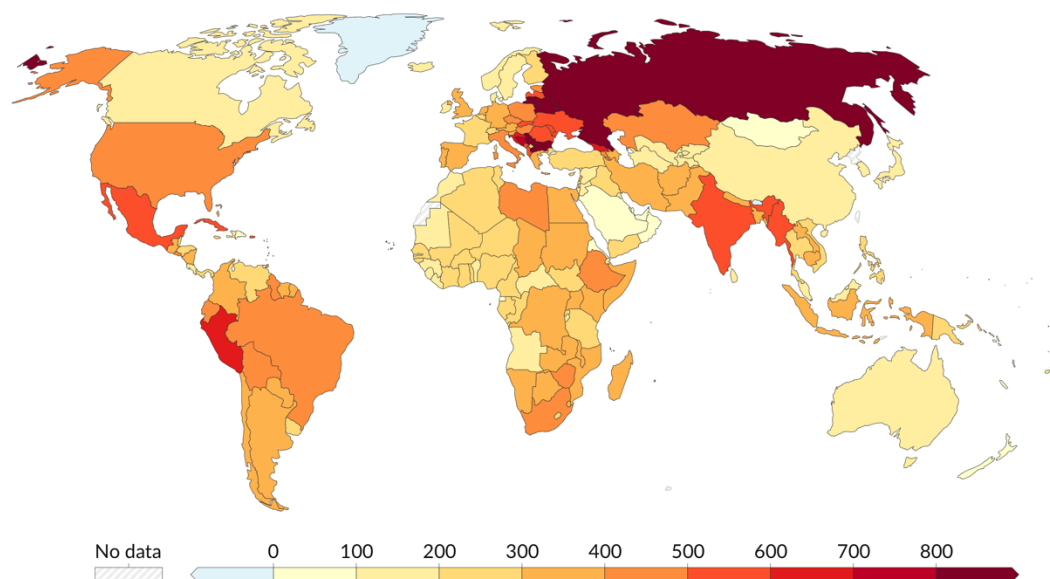
The COVID-19 pandemic has had a profound impact on the world, leading to widespread lockdowns and disruptions to social, economic, and environmental systems. Factories and airlines have been closed, public events have been postponed or canceled, schools and universities have switched to online learning, and supply shortages have occurred due to restrictions on transportation between countries. However, there have also been some positive outcomes from the pandemic, such as reductions in air pollution, greenhouse gas emissions, and water pollution.

According to the WHO's daily COVID-19 data, the COVID-19 situation in Asian countries is generally less severe than in European countries and China, which was the epicenter of the early outbreak. However, the number of cases in Asia is still significant, with over 300,000 infections reported to date. As of March 23, 2020, the global number of COVID-19 cases surpassed 332,930. As of today, COVID-19 cases have been confirmed in 17 countries and territories in the Western Pacific Region, 60 countries and territories in the European Region, 9 countries in the South-East Asia Region, 20 countries and territories in the Eastern Mediterranean Region, 45 countries and territories in the Region of the Americas, and 39 countries and territories in the African Region.

Estimated cumulative excess deaths per 100,000 people during COVID-19, Oct 29, 2023



For countries that have not reported all-cause mortality data for a given week, an estimate is shown, with uncertainty interval. If reported data is available, that value only is shown. On the map, only the central estimate is shown.



Data source:

CC BY

Note: For some countries, all-cause deaths and COVID-19 deaths use different date schemes, in which one refers to when the death occurred and the other to when it was reported. This difference could produce an artificial lag between the two time series.

Figure 1: Total Confirmed COVID-19 cases per 100,000 population, March 2020
Source: https://ourworldindata.org/explorers/coronavirus-data-explorer?time=latest&country=~OWID_WRL&Interval=Cumulative&Relative+to+Population=true&Color+by+test+positivity=false&Metric=Excess+mortality+%28estimates%29

Overall, the COVID-19 pandemic has had a profound impact on the world, both positive and negative. It is important to stay informed about the latest developments and to take steps to protect yourself and others from the virus

1.2.Problem Statement

This study aims to conduct a comprehensive GIS-based hotspot mapping and trend analysis of the COVID-19 pandemic on a global scale. The study is motivated by the need for a global perspective to understand the pandemic's multifaceted impact on populations, healthcare systems, and other organizations, beyond traditional epidemiological considerations.

The study will integrate remote sensing (RS) and geographic information system (GIS) technologies to create dynamic visual representations of the evolving spatiotemporal patterns of COVID-19 progression across continents and countries. Time-series analysis techniques will be used to identify significant trends and patterns. The study recognizes the importance of reliable and precise data collection over time and throughout geographical locations, acknowledging that careful consideration of both spatial and temporal factors is needed for meaningful insights.

The study is expected to provide valuable insights into the global impact of the COVID-19 pandemic, informing governments and healthcare professionals in making informed decisions about resource allocation, response strategies, and mitigation measures. Additionally, the study will contribute to our understanding of the use of RSGIS in epidemic management, providing a helpful framework for future research in this area.

1.3.Objectives:

- 1.Literature Review on developing Predictive Models and Trends Analysis
- 2.Literature Review on identifying high-risk areas by Hotspots

CHAPTER -2

2.1.Literature Review on Trend analysis of Covid-19 Data

This chapter will focus on the current and previous research works which are applying Trend analysis methods to get the research outcomes. Since trend analysis methods are widely popular in spatial analysis and it has variety of application areas. Therefore, research works on different analysis methods will be presented in this chapter.

2.1.1. Trend Analysis

([Healthknowledge](#)) Trend analysis is a fundamental component of epidemiological research, it offers excellent insights into the temporal dynamics of epidemics. This methodological approach involves the systematic inspection of patterns in disease occurrence over time. (Dornick et al., 2021) utilized trend analysis to identify specific temporal trends, such as the rise or fall in case numbers, and to develop predictive models based on historical data for predicting future epidemic or pandemic trends. (Tegally et al., 2023) The examination of these trends over time allows for the detection of shift patterns and the exploration of factors contributing to variations in epidemic dynamics. (Gebretensae & Asmelash, 2021) Trend analysis plays a key role in understanding the evolution of health crises and informing evidence-based strategies for prevention and response in the territory of public health, In this chapter we will discuss few methods on how trend analysis is done for epidemics

2.1.2. Trend Analysis by ARIMA

(Rajan Gupta, 2021) predicted an estimation model for the rising number of cases in regional and Indian states using time -series forecasting methods to predict the future trends.

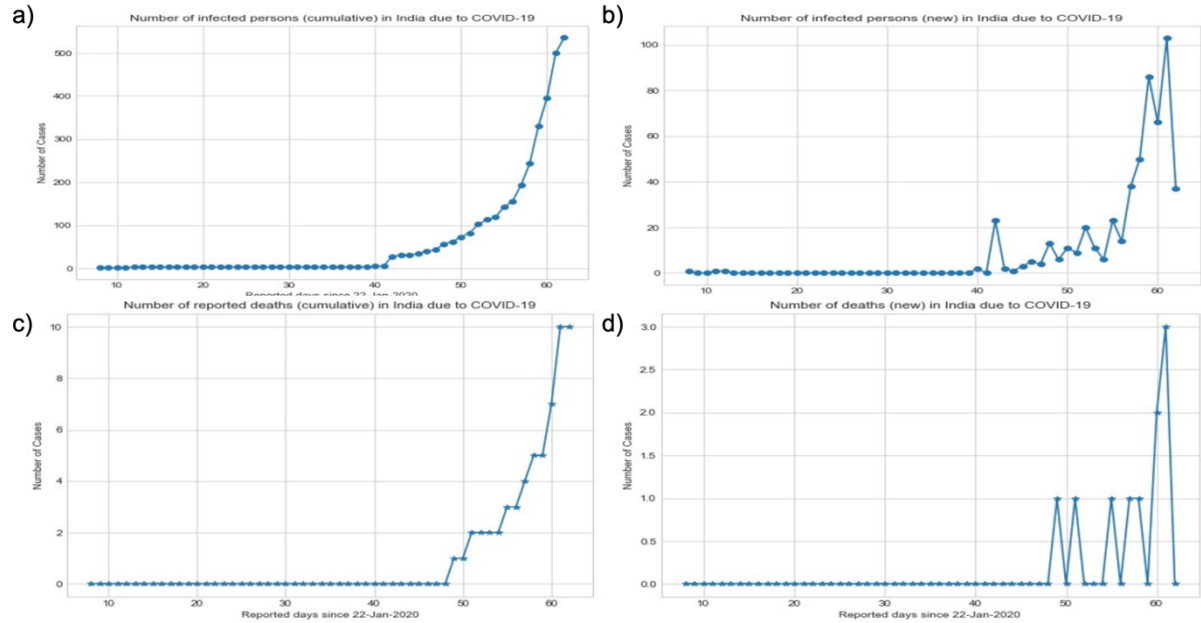


Figure 2: Trends for cases due to COVID-19 in INDIA a) Cumulative number of infected cases ,b) New Infected Cases ,c) Cumulative number of death cases , d) New Death Cases
 Source: <https://www.medrxiv.org/content/10.1101/2020.03.26.20044511v1>

Firstly, The trend analysis of number of infected cases and death cases has been assessed for India by using ARIMA modelling and Exponential Smoothing methods , (*Introduction to ARIMA Models*, n.d.) ARIMA(Auto Regressive Integrated Moving Average), A random variable that is a time series is stationary if its statistical properties are all constant over time. A stationary series has no trend, its variations around its mean have constant amplitude, and it wiggles in a consistent fashion, i.e., its short-term random time patterns always look the same in a statistical sense. A non-seasonal ARIMA model is classified as an ARIMA (p, d, q) model, where p is the number of autoregressive terms, d is the number of non-seasonal differences needed for stationarity, and q is the number of lagged forecast errors in the prediction equation. The forecasting equation is constructed as shown in eq. 1.

$$\hat{y}_t = \mu + \phi_1 y_{t-1} + \dots + \phi_p y_{t-p} - \theta_1 e_{t-1} - \dots - \theta_q e_{t-q} \quad (1)$$

Here the moving average parameters (θ 's) are defined so that their signs are negative in the equation, following the convention introduced by Box and Jenkins. For the current study, we have considered $p=1, d=1$ and $q=2$ to construct the ARIMA (1,1,2) model using equation 2, as these for these values the model's coefficients were computed to be most significant.

$$\hat{Y}_t = Y_{t-1} + \phi_1 (Y_{t-1} - Y_{t-2}) - \theta_1 e_{t-1} - \theta_2 e_{t-2} \quad (2)$$

Since the time series forecasting using ARIMA should be done on a stationary series, (Montañés & Sansó, 2001) The Dickey-Fuller test of stationarity was done to check the series. Rolling Mean and Rolling Standard Deviations were also calculated and the differences were calculated from the original data. The three methods used in the forecasting are all related to the (Kapusta et al., n.d.) Holt's Linear Exponential Smoothing methods. As per the exponential trends, the number of infected persons may grow very quickly in next 30 days and may reach to more than 80 million infected cases without any intervention.

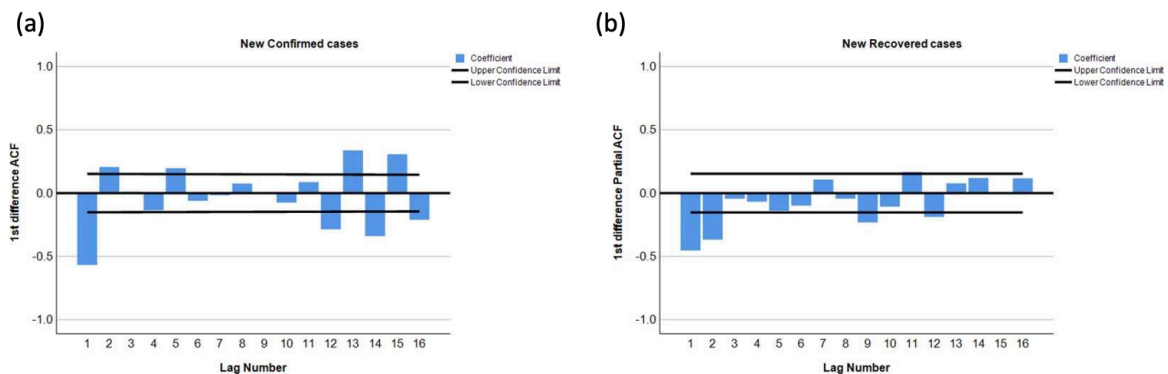


Figure 3: (a) Trend of cumulative confirmed cases (Natural Log) , (b) Trend of cumulative New Recovered cases

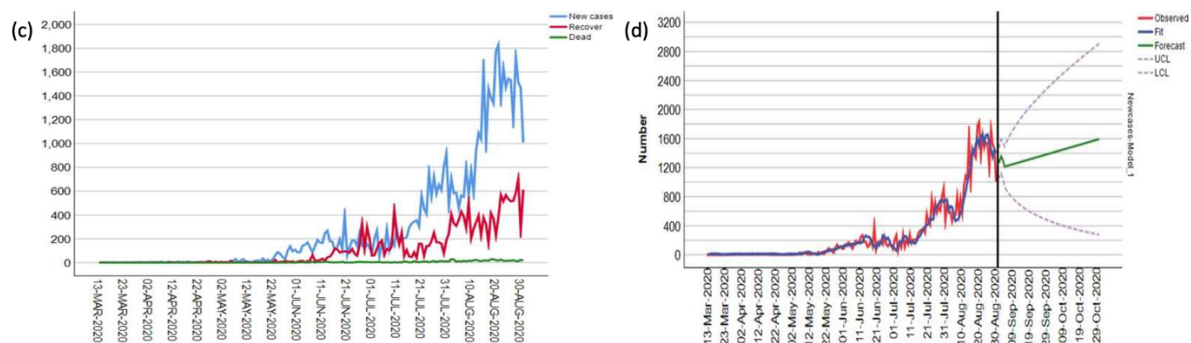


Figure 4: (c) Hot-Linear trend of total Confirmed cases of covid19, (d) A 60 days forecast of total confirmed cases of COVID-19 according to ARIMA models with 95% confidence interval

source: <https://www.jstor.org/stable/20076270>

(Gebretensae & Asmelash, 2021) ARIMA models were used to predict the progression of infectious diseases to identify the possible outcomes of an outbreak. However, artificial intelligence (AI) has the potential to help in all the stages of healthcare, from surveillance through to rapid diagnostic tests, and faster drug development. AI may also help to decide which patients should be prioritized for treatment and quickly learn which factors predict a higher risk of mortality, as well interventions and population-level controls, have led to reduced harm.

Date	Total Confirmed Cases Forecast Using ARIMA (0,1,5)	95% C.I for Total Confirmed Cases ARIMA (0,1,5)		Total Recovered Patients Forecast Using ARIMA (2,1,3)	95% C.I for TOTAL Recovered Patients ARIMA (2,1,3)	
		Lb95	Ub95		Lb95	Ub95
01-Sep-2020	1318	1106	1529	579	451	707
02-Sep-2020	1282	1069	1494	475	340	610
03-Sep-2020	1357	1123	1591	534	389	679
04-Sep-2020	1306	1057	1555	607	450	764
05-Sep-2020	1214	938	1490	510	346	674
06-Sep-2020	1221	895	1546	509	338	679
07-Sep-2020	1227	859	1595	613	432	794
08-Sep-2020	1234	828	1640	555	367	744
09-Sep-2020	1241	800	1682	498	305	691
10-Sep-2020	1248	774	1721	600	398	802
11-Sep-2020	1254	751	1758	599	390	809
12-Sep-2020	1261	729	1793	508	294	722
13-Sep-2020	1268	709	1827	575	355	796
14-Sep-2020	1275	690	1860	630	402	859
15-Sep-2020	1282	672	1891	537	303	770
16-Sep-2020	1288	655	1922	551	313	789
17-Sep-2020	1295	639	1952	642	396	888
18-Sep-2020	1302	623	1981	578	327	828
19-Sep-2020	1309	609	2009	537	283	792
20-Sep-2020	1315	595	2036	635	373	896
21-Sep-2020	1322	581	2063	620	353	887
22-Sep-2020	1329	568	2090	542	271	812
23-Sep-2020	1336	556	2116	615	339	891
24-Sep-2020	1343	544	2141	653	371	936
25-Sep-2020	1349	532	2166	565	279	851
26-Sep-2020	1356	521	2191	592	302	882
27-Sep-2020	1363	510	2215	670	373	966
28-Sep-2020	1370	500	2239	602	301	902
29-Sep-2020	1376	490	2263	577	273	881
30-Sep-2020	1383	480	2286	668	359	978
01-Oct-2020	1390	471	2309	642	328	956
02-Oct-2020	1397	462	2332	577	260	894
03-Oct-2020	1404	453	2354	653	330	975
04-Oct-2020	1410	444	2376	676	349	1004
05-Oct-2020	1417	436	2398	595	265	925
06-Oct-2020	1424	428	2420	632	298	966
07-Oct-2020	1431	420	2442	697	357	1036
08-Oct-2020	1437	412	2463	627	284	970
09-Oct-2020	1444	404	2484	616	270	962
10-Oct-2020	1451	397	2505	700	349	1051
11-Oct-2020	1458	390	2526	665	310	1020
12-Oct-2020	1465	383	2546	613	255	971
13-Oct-2020	1471	376	2567	689	327	1051
14-Oct-2020	1478	369	2587	700	333	1066
15-Oct-2020	1485	363	2607	626	257	996
16-Oct-2020	1492	357	2627	671	298	1044
17-Oct-2020	1498	350	2646	723	345	1100

(Continued)

Date	Total Confirmed Cases Forecast Using ARIMA (0,1,5)	95% C.I for Total Confirmed Cases ARIMA (0,1,5)		Total Recovered Patients Forecast Using ARIMA (2,1,3)	95% C.I for TOTAL Recovered Patients ARIMA (2,1,3)	
		Lb95	Ub95		Lb95	Ub95
18-Oct-2020	1505	344	2666	654	273	1034
19-Oct-2020	1512	339	2685	655	272	1039
20-Oct-2020	1519	333	2705	730	342	1119
21-Oct-2020	1526	327	2724	689	297	1080
22-Oct-2020	1532	322	2743	650	255	1044
23-Oct-2020	1539	316	2762	724	326	1123
24-Oct-2020	1546	311	2781	723	321	1125
25-Oct-2020	1553	306	2799	659	254	1064
26-Oct-2020	1559	301	2818	709	301	1118
27-Oct-2020	1566	296	2836	748	336	1161
28-Oct-2020	1573	291	2855	682	267	1097
29-Oct-2020	1580	286	2873	694	276	1112
30-Oct-2020	1587	282	2891	760	338	1182
31-Oct-2020	1593	277	2909	714	289	1139

Table 1:Forecasting of Daily Total COVID-19 Confirmed Cases and Total Recovered Patients in INDIA for the Next 60 Days
Source: <https://www.dovepress.com/trend-analysis-and-forecasting-the-spread-of-covid-19-pandemic-in-ethi-peer-reviewed-journal-of-healthcare>

2.1.3. Trend Analysis by Successive Approximation Method

(Qasim et al., 2020) predicted the daily reported cases and future state of the COVID-19 pandemic by utilizing Successive Approximation Method by using real-time data , (Republic of China, n.d.) By using the worldwide datasets with the Successive Approximation method, We define the two sets of data to find the mean ratio (η) of present cases count to the sum of previous and present cases. This ratio was further used to predict the future state of the expected cases and deaths counts. , the approximation produced can be further extended for the next day by subtracting the deaths (d) and approximating $x_i = \eta$ as

$$y_{n+1} = \frac{(y_n - d)(1 - \eta)}{\eta}$$

Where $\eta \times 0$, the entire population is expected to become infected similarly when $\eta \times 1$, the, the incidence of new patients counts significantly reduce.

$$y_{n+1} = \frac{y_n(1 - \eta)}{\eta}$$

The lower bound of proposed model is calculated by selecting $\eta = \max(x_i)$

By using this mean ratio (η), we have plotted the expected lower bound cases and maximum upper limit of our proposed model for the next 30 days from March 19th 2020 to predict the worldwide size of the pandemic in the coming days. It has been found that the lower bound count for confirmed cases for next 30 days will be 247007 and the maximum upper limit of cases will be 1667719

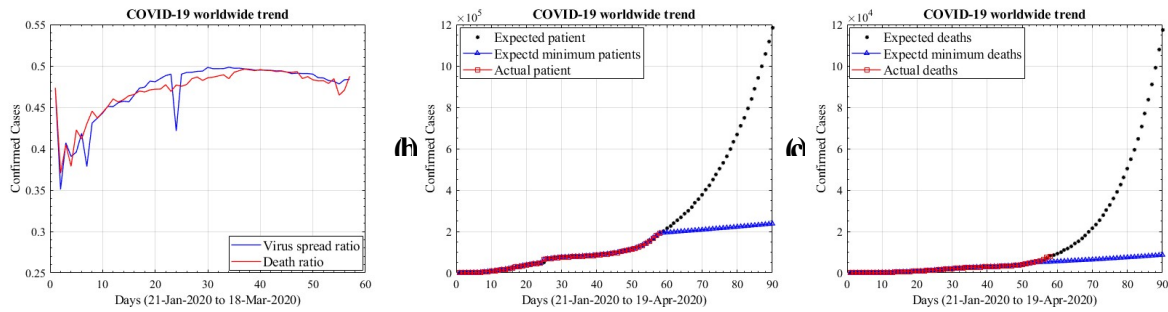


Figure 5: Worldwide trend of confirmed cases till March 19th 2020

Similarly, we evaluated and compared the countries strategies to contain COVID-19 based on their GHS index we plotted the pandemic trends for China, Japan, South Korea and Iran in order to assist other counties and international agencies to formulate their policies to address this pandemic, (Zhang et al., 2020) Our model showed that Japan successfully managed to contain the COVID-19 and stabilised the mean ratio (η) both for new confirmed cases and deaths despite having the earlier cases of COVID-19 and vast quantities of air traffic incoming from neighbouring countries, China, Republic of Korea, however according to the international media, the actual numbers of cases and deaths in Iran is much higher than official reports. Even with this available data, the set mean ratio (η), the maximum bound limit is indicating that the entire Iranian population is at increased risk of developing the infection.

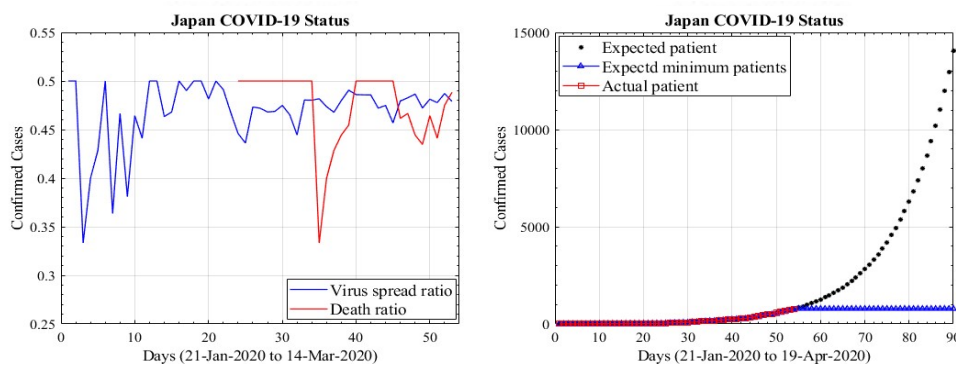


Figure 6: Mean Ratio and Prediction model of Japan

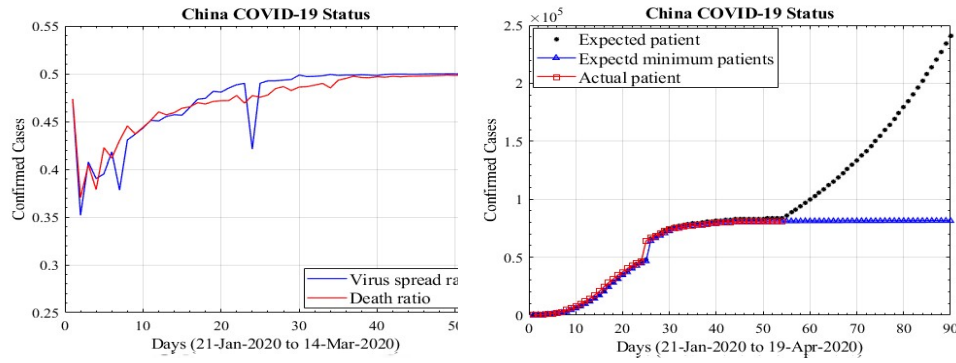


Figure 7: Mean Ratio and Prediction model of China

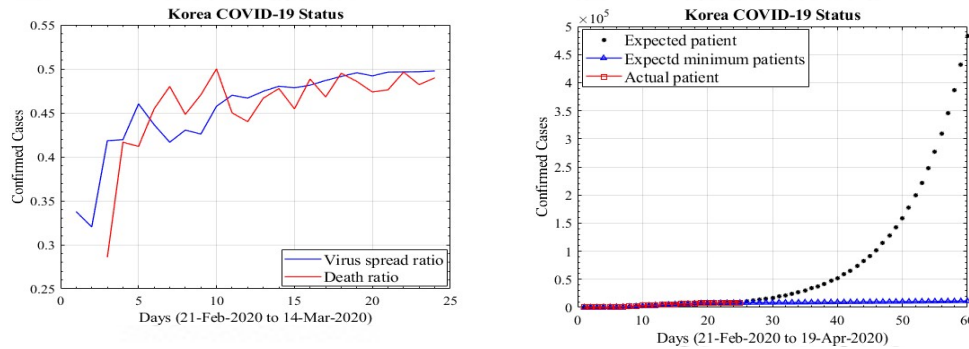


Figure 8: Mean Ratio and Prediction model of Korea

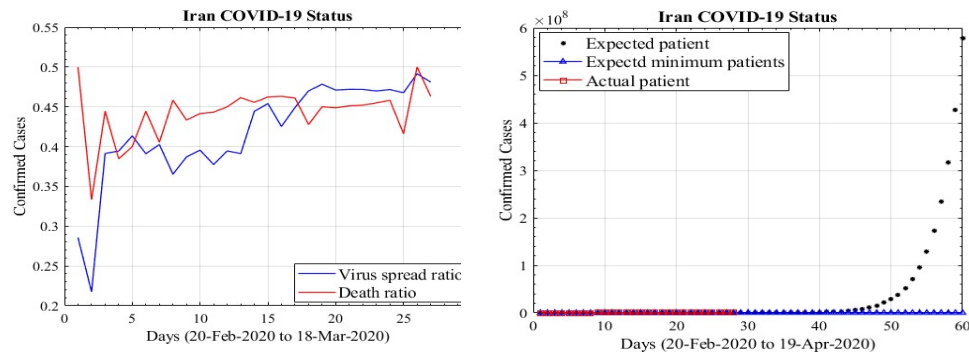


Figure 9: Mean Ratio and Prediction model of Iran

Source: <https://www.dovepress.com/trend-analysis-and-forecasting-the-spread-of-covid-19-pandemic-in-ethi-peer-reviewed-fulltext-article-IJGM>

2.2.3 : Trend analysis by Google Trend Analysis

(Pan et al., 2020) focused on predicting the number of confirmed Coronavirus cases utilizing different Google trend models like Linear model, Statistical model, ARIMAX, Generalised linear model (GLM), least absolute shrinkage and selection operator (LASSO) and Deep learning model. We have addressed that Google Trends data successfully predicted the

outcome in 33.3% of cases in 2008 and 39% in 2010. By analysing data from Google Trends, (Mavragani et al., 2018) calculated a valid approximation of the final result

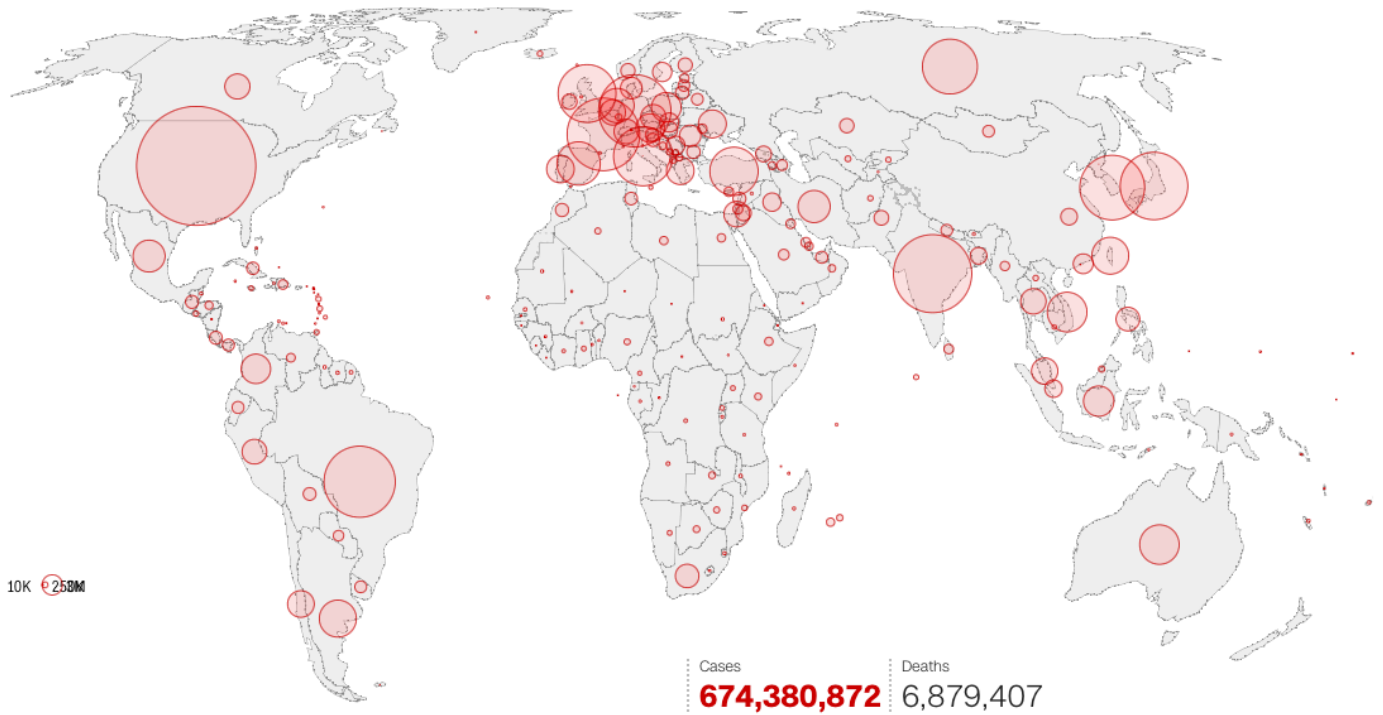


Figure 10: Coronavirus Confirmed Cases Worldwide(<https://edition.cnn.com/interactive/2020/health/coronavirus-maps-and-cases/>)

The Pearson correlation coefficient attempts to measure the similarity (correlation coefficient) between a series and the original one. Given two series X and Y , it can be defined as:

$$(X, Y) = \frac{\sum_{i=1}^T (x_i - \bar{x}) * (y_i - \bar{y})}{\sqrt{\sum_{i=1}^T (x_i - \bar{x})^2} * \sqrt{\sum_{i=1}^T (y_i - \bar{y})^2}}, \quad (1)$$

(Wang & Zhang, 2012). Multiple linear regression attempts to model the relationship between two or more explanatory variables and a response variable by fitting a linear equation to observed data. Essentially, it can be considered as an extension of ordinary least-squares regression that involves more than one explanatory variable. Suppose there are p distinct dependent variables, then the multiple linear regression model can be expressed as

$$Y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + \dots + \beta_p X_p + \varepsilon. \quad (2)$$

We took the Google Trend API to retrieve daily data from Jan 20, 2020 to March 23, 2020. The collection is done in both manner: a query for the data in a specific time range and a query for each day. Both of these types will produce trending of the search terms in a daily scale. %.

The deep neural network models use a dropout rate of 5% to avoid overfitting. The same strategy is applied for country based prediction.

	Correlation	p-value		Correlation	p-value
cases of covid19	0.8633	7.23E-19	coronavirus update	0.7796	2.17E-13
corona	0.7789	2.34E-13	covid	0.8650	5.14E-19
coronavirus	0.7408	1.32E-11	covid 19	0.8627	8.12E-19
coronavirus cases	0.8196	1.18E-15	covid 19 cases	0.8687	2.41E-19
coronavirus covid19	0.8174	1.62E-15	covid19	0.8506	7.91E-18
coronavirus news	0.7750	3.65E-13	covid19 cases	0.8584	1.87E-18
coronavirus symptoms	0.6664	6.18E-09			

Table 2:Correlation between Google Trend of search Queries and Confirmed cases Worldwide

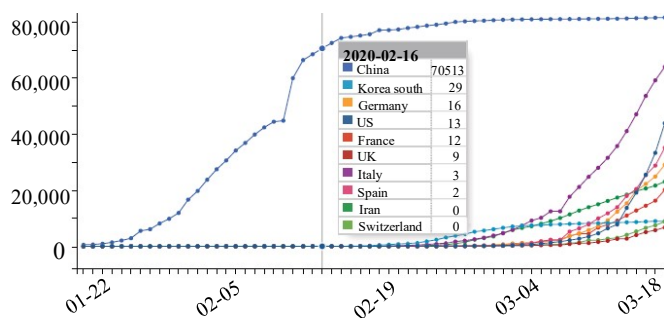


Figure 12:worldwide rise in Covid-19 source: www.Googletrends.co.se.11m24n/sfg

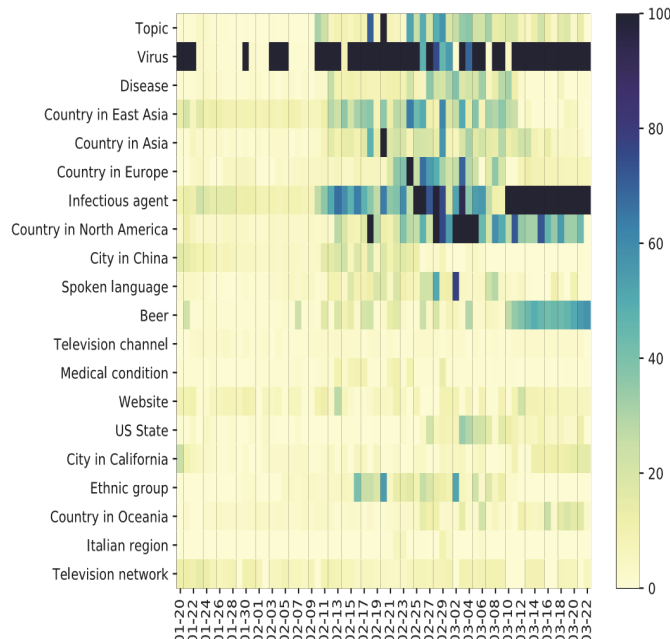


Figure 12:Broadcasting Of Covid-19 Queries
source: <https://www.cbsnews.com/news/coronavirus-corona-beer-they-have-nothing-to-do-with-each-other/>

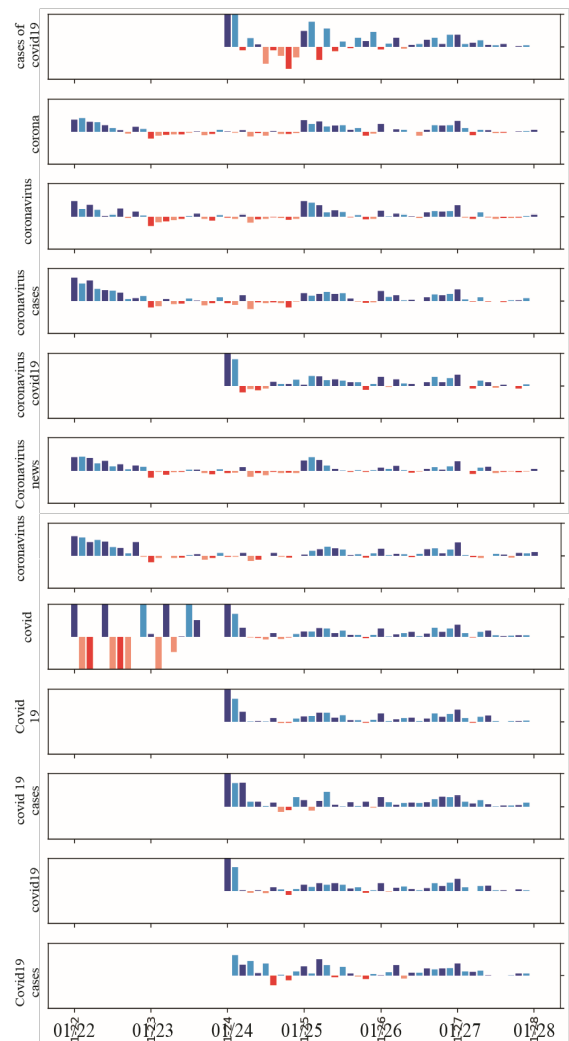


Figure 11: Evoulution of Internet Search Queries
source: <https://www.cdc.gov/coronavirus/2019-ncov/faq.html>

(Nguyen et al., 2020)The visualization of new cases prediction compared with actual values using the Deep Neural Network model. We can see that the Deep Neural Network model is able to predict the pattern and show a smaller gap

	RMSE	MAE	MAPE	R2
Linear	1,685	847	0.39	0.76
Linear + GT	1,683	1,475	1.43	0.38
Negative Binomial	1,645	1,194	1.74	0.77
Negative Binomial + GT	1,494	1,373	1.11	0.51
Deep Neural Network	1,595	1,317	0.48	0.76
Deep Neural Network + GT	807	569	0.63	0.82

Table 3: Confirmed Cases Forecast Performance by Learning Algorithms

2.2.4. Trend Analysis by Microsoft Software

(Liu et al., 2022) aimed to use the open and freely available data related to COVID-19 published online from around the world to explore changes in pandemic trends across six continents, and to provide valuable insight to better understand the epidemiological evolution of COVID-19. The data related to cases and death including before and after the SARS-CoV-2 variants from each country was acquired from ([OurWorldInData.org](https://ourworldindata.org)) and ([Gisaid.org](https://gisaid.org)), The data extracted and processed by month, including the number of cases, number of deaths, total cases per million, total deaths per million, and diversity profile of SARS-CoV-2 globally and across six continents; The acquired data were then cleaned and analysed using Microsoft Excel ([Microsoft Corporation](https://www.microsoft.com)), additionally the death rate was calculated as follows: death rate = deaths /confirmed cases $\times$ 100%. By March 31, 2022, a total of 490,071,097 cases of COVID

19 had been recorded, with 6,158,664 deaths globally (Figures 12(a) and (b)). The total cases per million and total deaths per million were 61,876.916 and 775.723, respectively

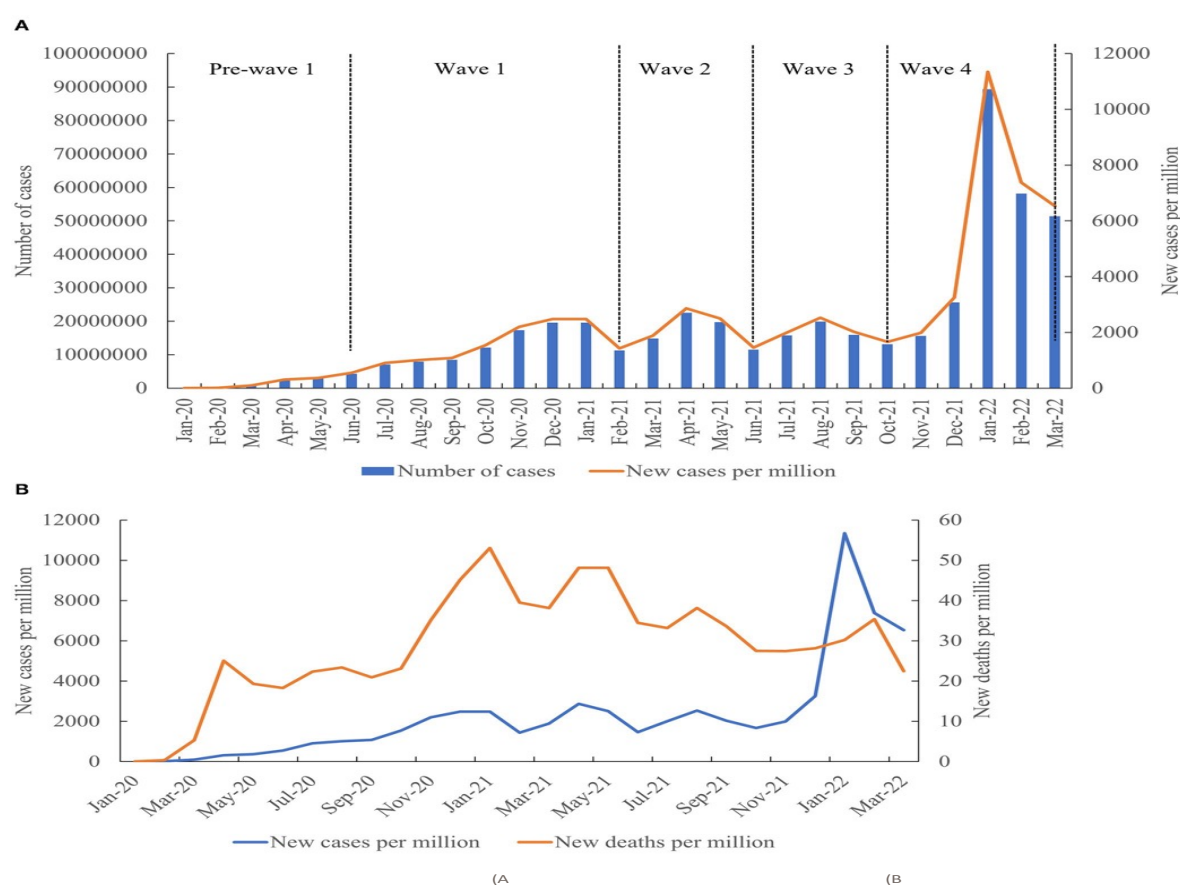


Figure 13: (a) Epidemiological trends in the six continents, (B) Evolution of cases and deaths per million in the six continents over time

Source: <https://www.frontiersin.org/articles/10.3389/fpubh.2022.957265/full#supplementary-material>

Figure 13(a) shows the number of deaths from low to high was 8,925 in Oceania, 251,740 in Africa, 1,264,662 in South America, 1,410,089 in North America, 1,450,615 in Asia, and 1,772,633 in Europe, Figure 13(b) shows the total deaths per million in other regions, from high to low, was 2,366.794 in Europe, 2,363.603 in North America, 310.065 in Asia, and 206.492 in Oceania, The highest incidence rate based on the cases per million was observed in Europe ($n = 240,656.542$), and the highest death rate based on the deaths per million was observed in South America ($n = 2,912.229$), The highest number of positive and fatal cases were recorded in Europe, whereas the lowest incidence was reported in Oceania

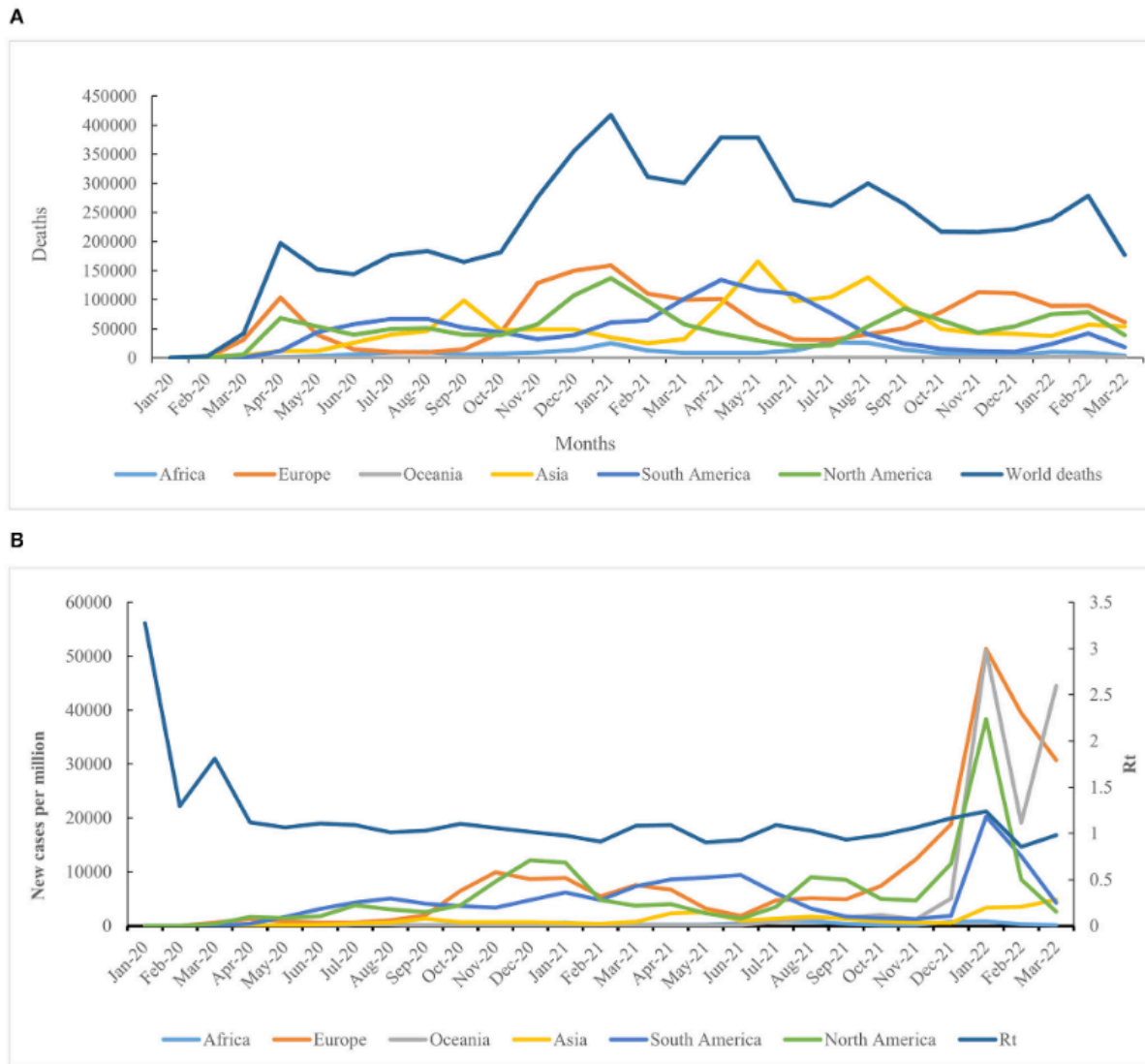


Figure 14:(a) Evolution of covid-19 related deaths in the six continents over time , (b) Dynamic fluctuation of the R_t -value in the six continents among the four waves of the COVID-19 pandemic
source: <https://www.frontiersin.org/articles/10.3389/fpubh.2022.957265/full>

2.2.5.Summary

This review presents a comprehensive overview of trend analysis methods and models applied to COVID-19 data. It explores the use of ARIMA models, the Successive Approximation Method, Google Trend Analysis, and Microsoft software for trend analysis and forecasting COVID-19 cases and deaths. This section highlights the significance of trend analysis in epidemiological research, offering insights into the temporal dynamics of epidemics. This section discusses the applications of trend analysis in predicting disease trends and informing

public health strategies. It provides a detailed analysis of each trend analysis method, discussing its strengths, limitations, and applications in the context of COVID-19. The review also compares the performance of different trend analysis models in forecasting COVID-19 cases and deaths. Overall, this review paper provides a valuable contribution to the field of trend analysis and its applications in the context of the COVID-19 pandemic. The review study offers insights into the evolving trends of COVID-19 and informs evidence-based public health interventions.

References	Methods	Application	Expected Results	Results
(Dornick et al, 2021)	NLP Evaluation (TFIDF, LDA, BERT, Sentiment Analysis)	COVID-19 Research Analysis	Discovering Research Topics, Analyzing Sentiment	Identifying General and Niche Themes, Neutral Sentiment
(Gebretensae et al,2021)	Box-Jenkins modeling and ARIMA method	Forecasting COVID-19 in Ethiopia	Identifying optimal ARIMA models, forecasting increasing cases	ARIMA (0, 1, 5) and ARIMA (2, 1, 3) selected, increasing cases forecasted
(Kapsuta,2020)	Linear Holt model for short-term predictions	Predicting COVID-19 cases in Ukraine	Accurate short-term forecasts	Challenges in Ukraine due to pandemic discussed
(Liu et al.2022)	Statistical Approach	Analyzing the epidemiological evolution of COVID-19 across six continents	Insights into pandemic trends and factors influencing COVID-19 spread	changes in confirmed cases and deaths, and the significance of viral variants

(Mavragani,2021)	Google Trends	Analyzing the methods, tools, and statistical approaches in Google Trends research	Provide an overview of Google Trends' usefulness and be a reference for future research	104 studies categorized by methodology, revealing the prevalence of time series analysis and the underutilization of Google Trends for predictions and forecasting in health assessment. Identified countries and languages of study. Seasonality often explored through visual evidence, cosinor analysis, ANOVA, and Kruskal-Wallis tests.
(Montañés and Sananó,2001)	Dickey-Fuller tests, seasonal pattern changes	Analyzing unit root null hypothesis in the presence of seasonal pattern breaks	tests tend to reject unit root hypothesis	Dickey-Fuller tests affected by the type and magnitude of seasonal pattern breaks, while Dickey-Hasza-Fuller test shows a different behavior.
(Nguyen et al,2020)	Correlation Analysis Between Google trends and Covid-19 cases	Investigating the relationship between Google search trends and COVID-19 spread to predict the number of cases.	To Show that Google search trends are highly associated with confirmed COVID-19 cases. Different regression models will be evaluated to understand the predictive power of Google Trends data.	Google search trends highly correlated with the number of reported COVID-19 cases. Deep Learning approach outperformed other forecasting techniques, demonstrating the potential of Google Trends data in predicting the spread of diseases.
(Ord and Getis,1992)	Getis Ord Static (Extending and exploring Gi and Gt statistics with nonbinary weights)	Detecting local spatial patterns in spatial-temporal data, e.g., AIDS epidemic near San Francisco	Improved spatial analysis with increased flexibility and nonbinary weight inclusion	Enhanced statistics for detecting spatial associations, distribution properties studied, and application in AIDS epidemic analysis discussed
(Qasim et al,2021)	Successive Approximation Method using global data	Modeling COVID-19 spread and death predictions	Mean ratios (η) for global cases and deaths	Mean ratio (η) for global cases: 0.485, deaths: 0.49; Worldwide lower bound cases: 247,007, maximum: 1,667,719; Deaths lower bound: 8,660, maximum: 117,397.

(RajanGupta,2020)	~Comparative analysis with SAARC nations. ~ARIMA (1,1,2) and Holt's LES models. ~Data from John Hopkins University (January 30, 2020, to March 24, 2020). ~Stationarity testing, rolling statistics, and log transformation. ~Python 3.7 and Jupyter Notebook.	Predicting COVID-19 trends in India	~Rising COVID-19 cases. ~Identification of most affected states. ~Projections for infected cases. ~Preparedness assessment.	Rapid COVID-19 case increase, regional disparities, and projection scenarios call for control measures.
(Wang and Zhang,2001)	Bayesian Quantile Regression, Simulation	Evaluating risk factors for violent crimes in the USA	Comparison of Bayesian quantile regression with simple regression and quantile regression, exploration of improper priors, and sensitivity analysis on parameter estimates	Uniform and normal priors yield similar results, which indicates that the improper uniform prior can work well when prior information is lacking
(Sheng and Zhang et al,2020)	The study estimated COVID-19 transmission on a cruise ship using R packages. R0 was estimated, and future cases were predicted.	~Estimated R0. ~Predicted daily new cases. ~Evaluated impact of R0 reduction.	R0 was estimated at 2.28. Daily cases were projected to increase, highlighting the need for infection control.	Early R0 estimation on the ship was crucial. Strict control measures were needed to prevent a large outbreak.

Table 4: Summary of Literature Reviews Presented for Trend Analysis

CHAPTER-3

3.1. Literature Review on Hotspot analysis on Covid-19 Data:

Hotspot analysis is a powerful tool for understanding the spatial distribution of COVID-19 cases and deaths, and for identifying areas at high risk of transmission. It can be used for a variety of purposes, including identifying areas at high risk of transmission, allocating resources, and monitoring the spread of the virus.

3.1.1. Hotspot analysis :

(Chow, J, 2016) Hotspot analysis is a spatial statistical technique that can be used to identify areas of high and low concentration of a particular phenomenon. (Feng, 2023) It helps pinpoint areas with the most severe disease outbreaks, enabling targeted allocation of resources such as testing kits and vaccines it tracks the spread of the epidemic over time, aiding in the development of containment strategies, like lockdowns or travel restrictions in rapidly spreading hotspot areas. (Lieberman et al., 2023) Hotspot analysis can also be used to track the spread of COVID-19 over time and to identify areas where the virus is spreading rapidly, it tracks the spread of the epidemic over time, aiding in the development of containment strategies, like lockdowns or travel restrictions in rapidly spreading hotspot areas. In the context of COVID-19, hotspot analysis can be used to identify areas where there is a higher concentration of COVID-19 cases, deaths, or other metrics

3.1.2. Hotspot Analysis by Spatial Correlation (Getis-Ord G_i^* Static)

(Parvin et al., 2021) conducted a geospatial analysis study to identify the Hotspot area of COVID-19, Getis-Ord G_i^* methodology adopted for detecting Covid-19 in substates of India, (Davis, 2016) Spatial correlation and autocorrelation are very important in spatial modelling, and various methods for testing and measuring spatial autocorrelation have been developed. Moran's I is one of the best-known spatial autocorrelation measures used by geographers, (Getis & Ord, 1992) However, it is often necessary to examine a pattern at a local scale, particularly if the process is spatially non-stationary. Thus, Getis and Ord focused upon local effects to develop a spatial association measure called the "Getis-Ord G_i^* statistic." Hotspot generates z-scores and p values. A significant hotspot is indicated by a high z-score and low p value, while a negative and low z-score indicates a significant cold spot. In the present study,

Getis-Ord G_i^* (G_i) was used to separate the clusters of high values (hotspot) from clusters of low values (cold spots). The Getis-Ord G_i^* statistics is expressed as:

$$G_i^* = \frac{\sum_{j=1}^n w_{ij} x_j - \bar{X} \sum_{j=1}^n w_{ij}}{S \sqrt{\left[\frac{n \sum_{j=1}^n w_{ij}^2 - \left(\sum_{j=1}^n w_{ij} \right)^2}{n-1} \right]}}$$

Equation 1: Getis-Ord Static

where y_k is the attribute value of the feature k ; $W_{l,k}$ is the spatial weight of l and k , n is the total number of features, and;;

$$\bar{X} = \frac{\sum_{k=1}^n y_k}{n}$$

$$S = \sqrt{\frac{\sum_{k=1}^n y_k^2}{n} - (\bar{X})^2}$$

Equation 2: High values and low values of G^* Static

The result expresses the high z-score and low p value of computed G_i^* which represent the statistical significance of spatial clusters

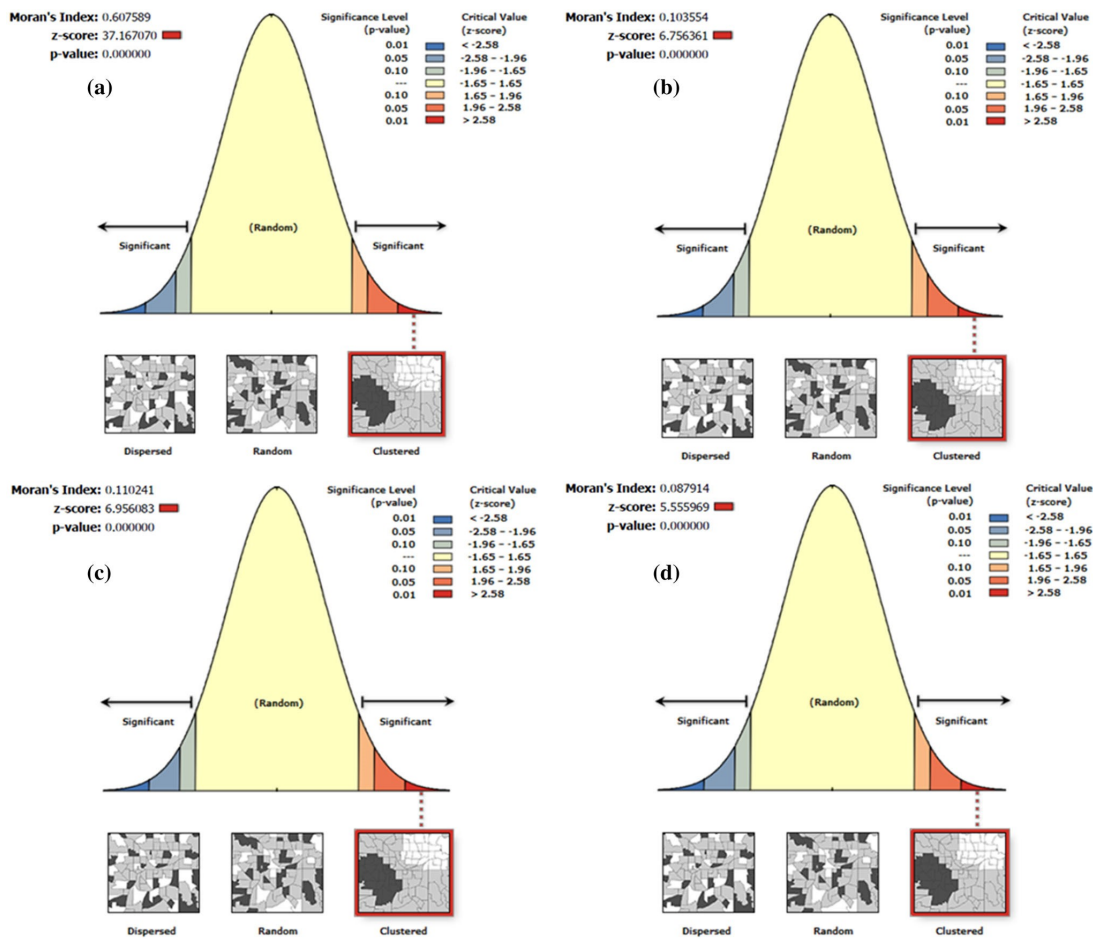


Figure 15: Graphical presentation of the result of spatial autocorrelation (a) total population, (b) population density, (c) foreign tourist arrivals and (d) confirmed cases of COVID-19 by using Moran's Index
source: <https://doi.org/10.1007/s41324-020-00375-1>

Factors	Observed (Moral's <i>I</i>)	Expected	Variance	z-score	<i>p</i> value	Pattern
Total population	0.6075	− 0.0013	0.000268	37.167	0	Clustered
Population denisty	0.1035	− 0.0013	0.000241	6.7563	0	Clustered
Foreign tourist arrivals (FTAs)	0.11	− 0.0013	0.000257	6.956	0	Clustered
Reported 2019-nCoV cases	0.0879	− 0.0013	0.000258	5.5559	0	Clustered

Table 5: Spatial autocorrelation of different factors considered for hotspot analysis of the COVID-19 using Moral's *I* statistics

The value of Moran's *I* index (observed) ranges between 0.1 and - 1. The indicates statistically significance of the cluster positive values imply more clustering and significant. $p \setminus 0.01$

(Shariati et al., 2020) Based on district-wise reported COVID-19 cases, the result showed that excepting the districts of the north-eastern state of India and Union territory (UT) like Puducherry, Andaman & Nicobar Islands, Lakshadweep, Daman Diu, and Dadra & Nagar Haveli, more or fewer districts of all others states are hotspot to the COVID-19 with a high value of z-score (<0.8745), These were least considered due to lower z-score of Getis-Ord G_i^* statistic.

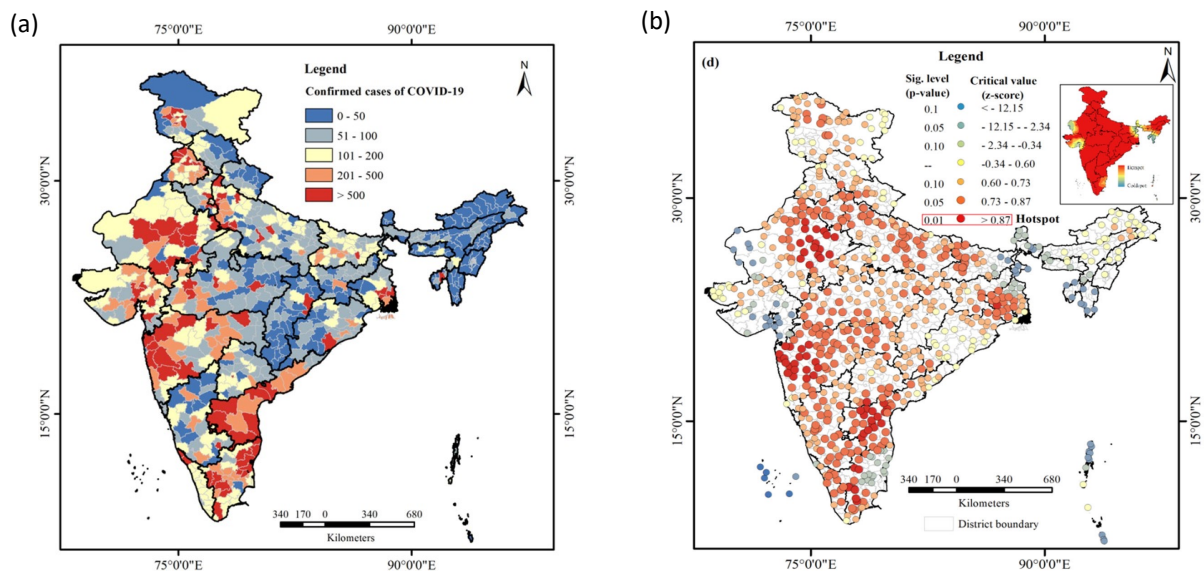


Figure 16: (a) District wise confirmed cases of covid-19, (b) Hotspot map of India based on District wise confirmed cases
source: <https://doi.org/10.1007/s41324-020-00375-1>

3.1.3 Hotspot Analysis by Machine Learning Methods (Gaussian Mixture model)

(Lieberman et al., 2023) aimed to provide a data-driven methodology and comprehensive case study to uncover location specific virus dynamics within a given area. (Yufeng, 2021) The methodology uses an unsupervised Gaussian Mixture model to cluster cases at a desired granularity. (Pedregosa et al., 2018) Unsupervised machine learning method(Gaussian mixture models) This model allows us to group cases based on their location. The output clusters can therefore be used for analyse and to model the dynamics of the virus within the determined area. The generation of the Gaussian Mixture Model distributions and corresponding HTML maps were implemented in Python 2.7, utilising the Sckit-learn API package , This is combined

with an epidemiological analysis to quantify each cluster's progression and whether it can be defined as a hot-spot and non-hotspot clusters, this analytical framework is applied to the first and second waves, taking Gauteng province, South Africa, as a case study. (Dempster et al., 2018) A Gaussian Mixture model is an algorithm which operates by generating k 2-dimensional Gaussian probability distributions, where k is a specified hyper-parameter. Thus, we are required to generate means, μ_j , covariance Σ_j and weighting, ϕ_j , where the index specifies the j Gaussian cluster. So, the probability of a new case, $p(x)$, occurring at a given point x is a linear combination of probabilities from all the generated clusters:

$$p(x) = \sum_{j=1}^k \phi_j N(x|\mu_j, \Sigma_j), \quad (1)$$

$$\gamma_{ij} = \frac{\phi_j N(x_i|\mu_j, \Sigma_j)}{\sum_{q=1}^k \phi_q N(x_i|\mu_q, \Sigma_q)} \quad (2)$$

$$\phi_j = \sum_i \frac{\gamma_{ij}}{N}, \mu_j = \sum_i \frac{\sum_i \gamma_{ij} x_i}{\sum_i \gamma_{ij}}, \Sigma_j^2 = \frac{\sum_i \gamma_{ij} (x_i - \mu_j)(x_i - \mu_j)^T}{\sum_i \gamma_{ij}} \quad (3)$$

Equation 3: Gaussian for new mean μ_j , covariance Σ_j and weight ϕ_j

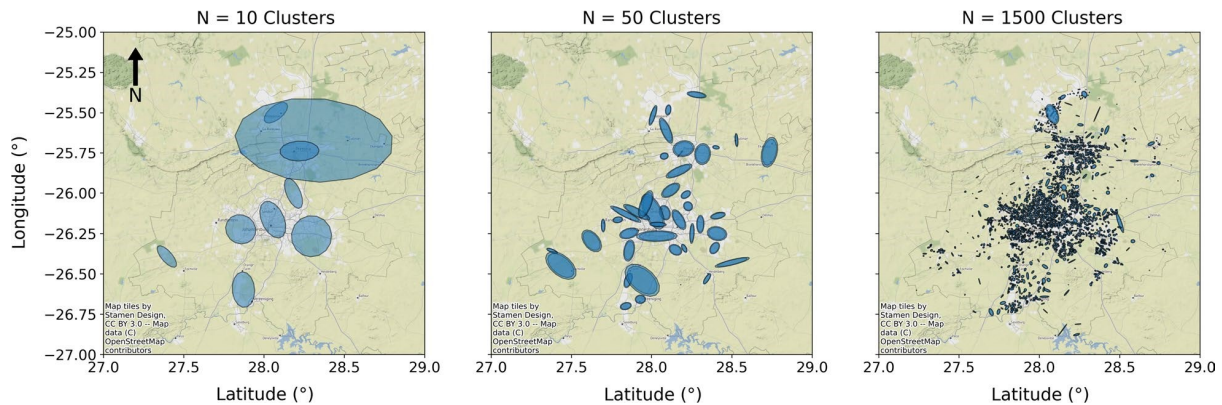


Figure 17: Visualisation of GMM cluster output for different number of clusters. (a) 10 Clusters. (b) 50 Clusters. (c) 1500 Clusters Source: <https://doi.org/10.1186/s12911-023-02098-3>

After applying the GMM algorithm for the clustering of cases in Gauteng, the number of clusters generated is an important metric. The number of clusters must be selected to best describe the specific virus dynamics, with at least an average of 100 cases per cluster. It was determined that 1500 clusters was the optimal number of clusters for the Gauteng province.

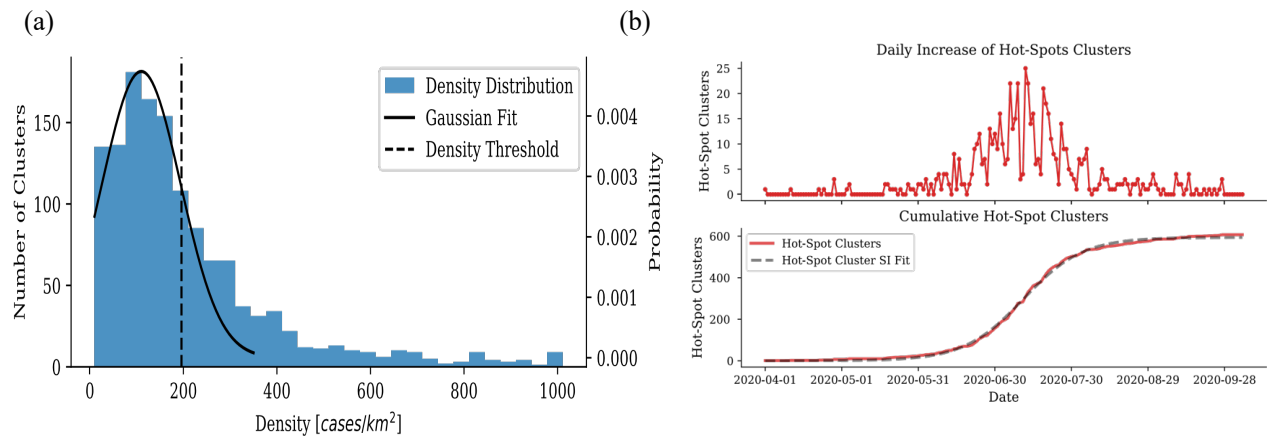


Figure 18: (a) first wave hotspot clustering, (b) second wave increase in hotspot clustering Source: <https://doi.org/10.1186/s12911-023-02098-3>

Since epidemics are situations which are highly changeable and constantly under flux, we can anticipate that hot-spot analysis can aid stakeholders in making informed, evidence-based and data-driven decisions, during epidemic waves and efforts such as vaccine roll-outs.

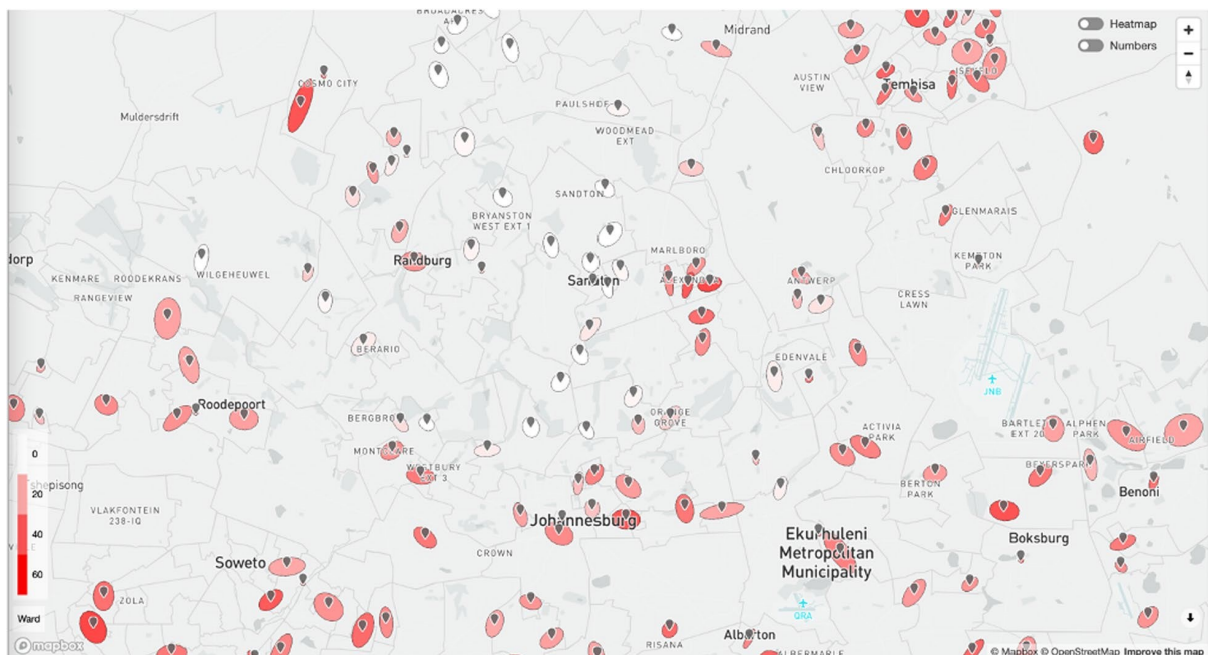


Figure 19: Hot-spot visualisation. Hot-Spot visualisation on gpcoronavirus.co.za. Courtesy of IBM South Africa Source: <https://sacoronavirus.co.za/covid-19-risk-adjusted-strategy/>

3.1.3. Hotspot Analysis by Kernel Density Estimation :

(Xu et al., 2023) launched a High-Resolution Hong Kong COVID-19 hotspot map by using Kernel Density Estimation model combining with Machine learning and Python models for graphical and Hotspots interpretation, (*High-Resolution Hong Kong COVID-19 Hotspot Map Launched*, 2022) provides a visualisation of real-time and dynamic geographic distribution of covid-19 cases in the city, it was developed by Hong Kong Baptist University, (Kato, 2021) the colour visualisation of COVID-19 risk levels on the Map is based on the automatic, real-time updating and computation of infection case data. To meet the Map's functional requirements, the updating and computation process must be done timely, and support the creation of high-resolution images. A computational tool for spatial-temporal data analysis named Kernel Density Visualisation (KDV) is used to support the visual creation, For the Space Time kernel Density Estimation (STKDE) analysis, (Nakaya, 2020) developed the Space-Time Density Tool for ArcGIS Pro and made it available to the public. This program was used in the analysis of this study to estimate the kernel density from the LH data and create an iso-surface. In the analysis, the density values of the iso-surface were set to 95%, 99%, and 99.9%, meaning that in this study, iso-surfaces were created for the top 5%, 1%, and 0.1% of the density distribution.

$$\hat{f}(x, y, t) = \frac{1}{nb_s^2b_t} \sum_{i=0}^n K_s\left(\frac{x-x_i}{b_s}, \frac{y-y_i}{b_s}\right) K_t\left(\frac{t-t_i}{b_t}\right) \quad (1)$$

$$K_s(u, v) = \begin{cases} \frac{2}{\pi}(1 - (u^2 + v^2)), & (u^2 + v^2) < 1 \\ 0, & otherwise \end{cases} \quad (2)$$

$$K_t(w) = \begin{cases} \frac{3}{4}(1 - w^2), & w^2 < 1 \\ 0, & otherwise \end{cases} \quad (3)$$

Equation 4: Space-time kernal density equation

where $\hat{f}(x, y, t)$ = the density estimation at the location (x, y, t), n is the number of points, b_s is the spatial bandwidths, b_t is the temporal bandwidths, u is the difference rate between longitude x_i and $x_{i+1} + 1$, v is the difference rate between latitude y_i and $y_{i+1} + 1$, w is the difference rate between the time t_i and $t_{(i+1)} + 1$, (HKBU-COVID-19 hotspot, 2021) while conventional KDV can be supported by many statistics, geographical and visualisation software tools, none of them can support high-resolution KDV with million-scale datasets as required by the Map. Furthermore, computation methods of conventional KDV cannot support the real-time creation of dynamic images

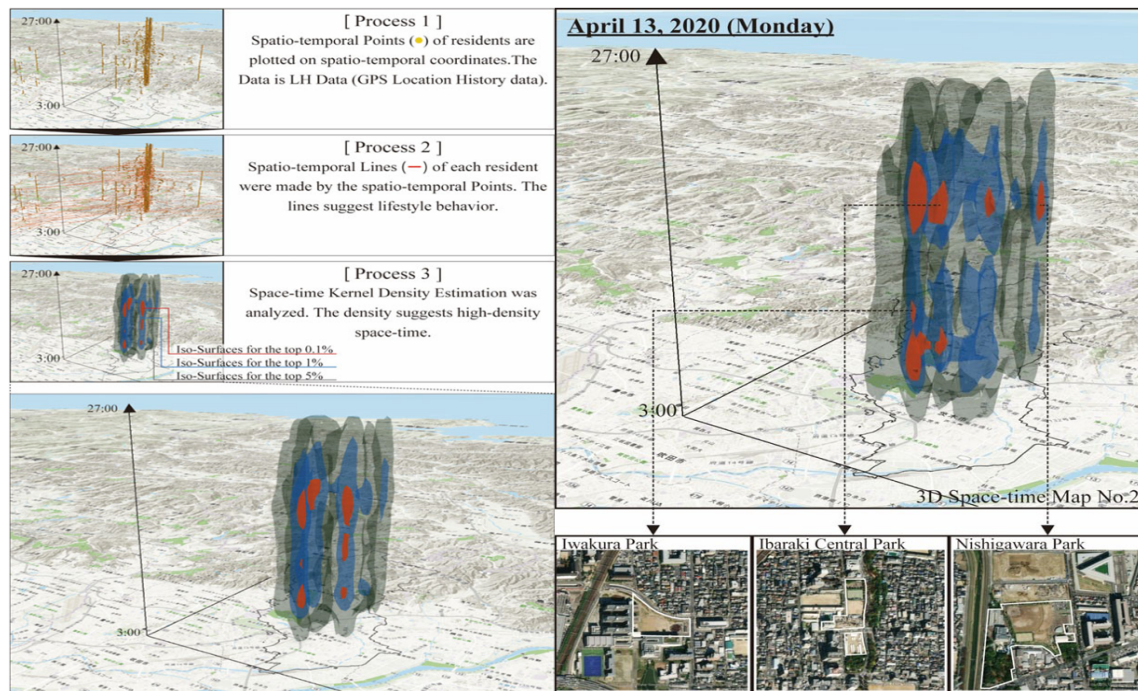


Figure 20: Visualisation of STKDE with LH data of Covid-19 in 2020
source: https://link.springer.com/10.1007/978-3-030-76059-5_4

(Byron CHOI — Hong Kong Baptist University, n.d.) the research team developed new computation methods with the application of complexity-optimised theory. The new computation methods work together with a progressive visualisation framework to create continuous partial visualisation outputs, Using the new computation methods, resolution of the Hong Kong COVID-19 Hotspot Map can be increased to 1376 x 960 pixels (high definition resolution) and process one million data points, and a response time of less than 0.5 second can be achieved which substantially out-performs other existing tools.

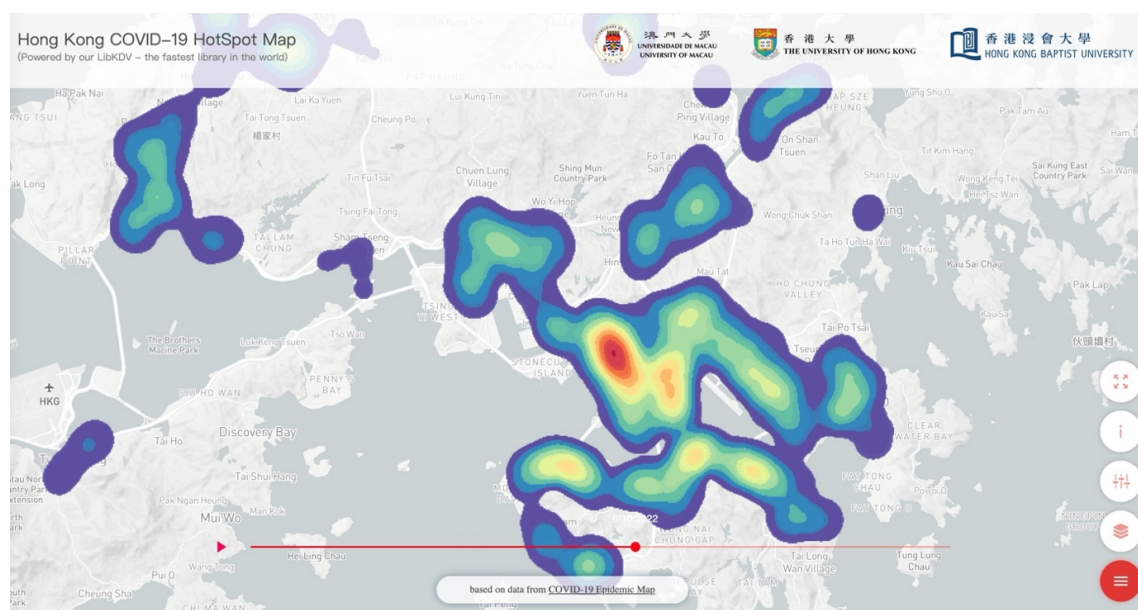


Figure 21: High-resolution Hong Kong COVID-19 Hotspot Map Source: <https://www.hkbu.edu.hk/en/whats-new/press-release/2022/hkbu-led-research-team-launches-hong-kong-covid-19-hotspot-map.html>

3.1.4.Hotspot Analysis by Open Source Data:

(Foraker, 2022) describes the methods of building a real-time visualization of patients with COVID-19–positive tests, which facilitates timely public health response to the pandemic, (*ArcGIS Solutions | Documentation*, n.d.) ArcGIS Hub site used by public health agencies to communicate vaccination plans, increase public confidence in COVID-19 vaccinations, and ensure the public knows when, and where, they can be vaccinated, as it had developed an interactive street-level visualization that shows total confirmed cases over time and resolving after the occurrence of infection. Our source data included patient demographics (ie, age, race and ethnicity, and sex), street address of residence, respiratory test results, and date of test. To find out the active cases we used the open available source data from (*World Health Organization (WHO)*) and (Johns Hopkins Coronavirus Resource Center, 2023) and put them into an .csv file to understand the total active cases as

$$\begin{aligned} \text{Active cases} = & 100\% \text{ of new cases in past 14 days} + 19\% \text{ from past 30days} \\ & + 5\% \text{ from past 56days} - \text{total deaths} \end{aligned}$$

Equation 5:Equation for Total Active confirmed cases

Each dot corresponded to a positive case. Fourteen days after the date of the respiratory test, we considered the case resolved and removed the dot from the visualization. date. The results shows where new cases developed in the region and how patterns changed over the course of the pandemic

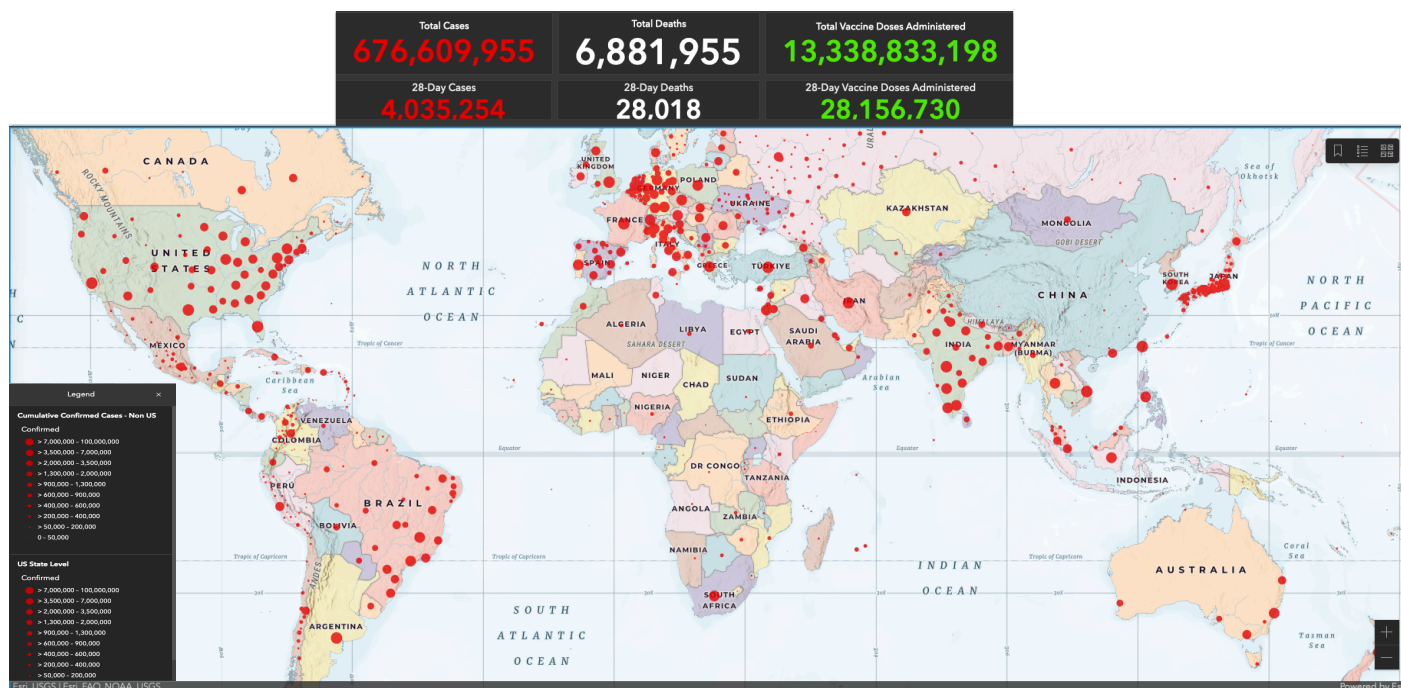


Figure 22: The US, India and Brazil have seen the highest number of confirmed cases, followed by France, Germany and the UK.
source: <https://coronavirus.jhu.edu/map.html>

(*ESRI-Coronavirus COVID-19 Cases*) this features layer contains the most up-to-date COVID-19 cases and the latest trend plot. It covers the US (county or state level), China, Canada, Australia (province/state level), and the rest of the world (country/region level). On March 10, 2023, the Johns Hopkins Coronavirus Resource Center ceased collecting and reporting global COVID-19 data, Similarly, (Care, 2023) found out the total number of vaccinated doses from first wave and second wave of covid-19, health.gov.au an open source government data provided the complete data from 19 March 2019 to 27 October 2023

Country	Percentage of population vaccinated	Additional Doses	Doses administered	Fully vaccinated per 100 people	Total Vaccinated
World	72%		35	2,71,64,91,503	13,27,01,70,332
Macau	>99%		76	4,77,572	17,97,836
Brunei	>99%*		78*	340,030*	1,289,036*
U.A.E.	>99%*		53*	5,138,699*	24,922,054*

Qatar	>99%		67	19,03,612	76,06,840
Samoa	97%*	90%*	42*	82,117*	452,831*
Nepal	97%	85%	32	90,25,399	6,09,76,433
Chile	95%*	93%*	145*	27,474,929*	62,688,847*
Malta	95%	94%	92	4,64,470	13,80,226
Portugal	95%*	87%*	69*	7,043,690*	28,106,464*
Nicaragua	95%	92%	-	-	1,22,41,350
Cuba	95%	88%	77	87,01,858	4,36,38,303
Vietnam	94%*	89%*	60*	57,743,764*	266,211,063*
China	94%*	91%*	59*	826,913,000*	3,491,077,000*
Peru	93%	88%	89	2,89,42,497	8,77,82,114
Hong Kong	93%	91%	94	69,71,874	2,06,81,138
Bangladesh	93%	84%	41	6,73,37,563	35,49,33,184
Cambodia	93%	89%	64	1,05,63,979	4,61,18,633
Canada	92%*	84%*	81*	30,431,178*	96,956,502*
Argentina	92%	78%	79	3,56,81,131	11,48,21,208
Taiwan	92%	87%	103	2,43,99,397	6,69,93,502
Bhutan	92%*	89%*	83*	634,641*	2,011,426*
Costa Rica	91%*	85%*	56*	2,832,234*	12,778,229*
Singapore	90%*	90%*	78*	4,440,289*	14,727,569*
Brazil	90%	83%	60	12,56,08,160	48,49,95,271
Mauritius	89%	86%	51	6,39,939	26,14,136
Ecuador	88%	82%	60	1,04,89,626	3,94,85,025
Malaysia	88%	86%	54	1,71,21,011	7,25,74,547
Spain	88%	87%	56	2,65,17,510	10,55,22,606
Australia	88%*	85%*	78*	19,790,466*	63,681,652*
New Zealand	87%*	84%*	72*	3,536,866*	11,977,450*
Uruguay	87%	84%	88	30,34,683	89,39,862
South Korea	87%	86%	80*	41,325,954*	129,647,782*
Kiribati	86%	69%	32	37,606	2,19,980

Iceland	86%*	80%*	69*	250,542*	805,469*
Colombia	85%*	73%*	29*	14,675,916*	90,210,929*
Aruba	85%	79%	-	-	1,74,388
Italy	84%	80%	78	4,69,06,547	14,37,48,677
Botswana	84%	71%	22	5,03,402	31,71,667
Rwanda	84%*	78%*	47*	5,879,013*	26,106,436*
Tonga	84%	74%	37	38,447	2,02,720
Ireland	83%	82%	64	31,83,819	1,11,11,161
Panama	83%*	75%*	48*	2,027,869*	8,756,680*
Japan	83%	82%	138	17,37,06,902	38,16,75,627
Denmark	82%	82%	63	36,52,503	1,31,45,231
Kuwait	82%	80%	35	14,57,469	82,58,651
Laos	82%*	73%*	-	-	11,111,066*
Finland	82%	79%	78	43,24,621	1,31,96,843
Thailand	82%*	77%*	46*	32,143,431*	142,635,014*
France	82%	79%	71	4,75,53,764	15,42,95,635
Norway	81%*	76%*	70*	3,743,839*	12,145,560*
Belgium	81%	80%	101	1,16,19,781	2,96,22,669
U.K.	81%*	76%*	60*	40,373,987*	151,248,820*
Fiji	80%	72%	19	1,70,903	15,52,809
United States	80%*	68%*	34*	111,367,843*	632,905,664*
Liberia	79%*	76%*	0*	1,987*	4,460,668*
Saudi Arabia	79%*	74%*	31*	10,793,985*	68,148,406*
Sri Lanka	79%*	68%*	38*	8,220,002*	40,116,590*
Iran	79%	71%	38	3,13,52,288	15,77,85,811
Germany	78%	76%	78	6,47,26,123	19,21,02,093
Luxembourg	78%	75%*	74	4,58,672	13,74,048
Venezuela	78%*	52%*	2*	651,502*	37,860,994*
Austria	78%*	77%*	76*	6,733,783*	20,263,306*
Mexico	78%*	65%*	45*	57,026,630*	225,063,079*

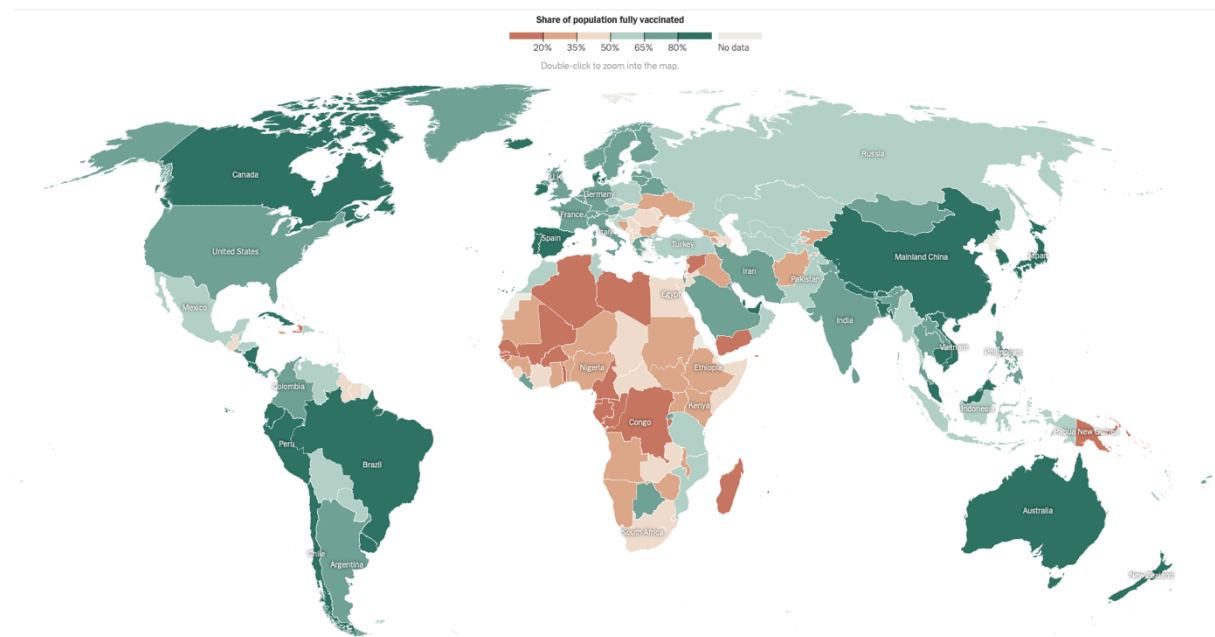
Sweden	76%	74%	96	98,93,062	2,56,64,081
Bahrain	76%*	75%*	61*	1,008,663*	3,476,633*
Indonesia	75%*	64%*	25*	67,952,274*	444,303,130*
Maldives	75%*	73%*	31*	167,187*	951,511*
India	75%	70%	17	22,70,99,567	2,20,63,48,628
Pakistan	75%	64%	23	5,01,14,329	33,36,62,084
Israel	74%	68%	64	57,60,855	1,86,42,215
Greece	74%	71%	66	71,09,680	2,21,09,103
Netherlands	74%*	69%*	54*	9,417,108*	36,104,942*
Philippines	73%	68%	20	2,16,42,800	17,04,57,489
El Salvador	72%*	67%*	36*	2,292,292*	11,289,175*
Switzerland	71%	70%	57	48,76,829	1,69,26,060
Mongolia	70%*	67%*	32*	1,044,337*	5,492,919*

Table 5: Countries with total no of vaccinated population

source: <https://www.health.gov.au/topics/covid-19/weekly-reporting?language=und>

(Holder, 2021) uses the same data to track coronavirus vaccinations around the world , and concludes more than 5.55 billion people worldwide have received a dose of a Covid-19 vaccine, equal to about 72.3 percent of the world population

Figure23:Percentage of fully vaccinated population([nytimes.com](https://www.nytimes.com))



3.1.5.Summary.

Hotspot analysis is a valuable approach in spatial epidemiology and disease tracking, particularly during a dynamic and evolving epidemic like COVID-19. This research chapter has explored various hotspot analysis methods, including Hotspot Analysis, Hotspot Analysis by Spatial Correlation (Getis-Ord G_i^* Static), Hotspot Analysis by Machine Learning Methods (Gaussian Mixture model), and Hotspot Analysis by Kernel Density Estimation., Getis-Ord G_i^* statistics have proven effective in identifying areas with high concentrations of COVID-19 cases, while Kernel Density Estimation in combination with machine learning models has enabled the creation of high-resolution hotspot maps that provide real-time insights into the geographic distribution of cases. Additionally, open-source data from the World Health Organization and the Johns Hopkins Coronavirus Resource Centre has been utilized to monitor and visualize active COVID-19 cases, facilitating public health response and vaccination plans. Overall, hotspot analysis methods are essential tools for decision-makers and epidemiologists to make informed, data-driven decisions to mitigate the spread of the virus and allocate resources effectively.

References	Methods	Application	Expected Results	Results
(Davis,2016)	Statistics and Data Analysis in Geology	Clarification of the zonation procedure	Clarifying the zonation procedure for geology data analysis	Explanation of standardized values, total sum of squares, and squared distances for the zonation process
(Dempster et al,2018)	Algorithm for computing maximum likelihood estimates from incomplete data	Including missing value situations, grouped data, and factor analysis	Monotone likelihood behaviour and algorithm convergence	Demonstrating its applications presented at different levels of generality
(Zhiqiang and Fenf,2021)	Spatiotemporal patterns were analyzed using SaTScan, and geographically weighted Poisson regression (GWPR)	To investigate the association with socioeconomic and environmental factors in England for First and second waves of the pandemic	Identifying significant spatiotemporal variation in COVID-19 deaths, and exploring the association between mortality	The GWPR analysis revealed consistent associations with factors such as age composition, ethnic composition, deprivation, care home availability, and pollution.
HongKong Baptist University(HKBU)	Kernel Density Visualization and Kernel Density estimation model	Real-time visualization of COVID-19 cases in Hong Kong, monitoring infection risks	Enhanced data resolution and speed in visualizing COVID-19 cases	It provided high-resolution images, real-time dynamic visualizations, and a response time of less than 0.5 seconds
(Kato and Haruka,2020)	Space-time kernel density estimation	Visualizing space-time with high population density to help local governments designate specific areas for restrictions	~Evaluation of the effectiveness of STKDE in understanding high-density space-time. ~Insight into the impact of the Japanese soft lockdown on population density and daily lives.	~Designation of Specific Space time Areas, Effectiveness of early Preventive measures, Alternative to soft lockdowns.

(Lieberman et al ,2023)	~Gaussian Mixture Model, Analysis of cluster's severity and progression	~Analysing the dynamics of COVID-19 in Gauteng Province, South Africa ~Providing data-driven methodology to expose location-specific virus dynamics	~Identification and classification of virus hot-spots ~Temporal progression and severity assessment of these clusters ~Improved understanding of the virus's spread for informed local interventions	Aims to help policy makers and local populations understand the virus dynamics for localized interventions.
(Parvin et al,2021)	Geospatial analysis, spatial autocorrelation, Getis-Ord Gi* statistics	Identifying COVID-19 hotspots in India	High-risk areas based on total population, population density, foreign tourist arrivals, and confirmed COVID-19 cases	Hotspots identified in Maharashtra, Tamil Nadu, Gujarat, Delhi, Uttar Pradesh, and West Bengal based on total population and reported COVID-19 cases. The geospatial methodology aids in implementing strategies to combat the spread of COVID-19.
(Mohsen Shariati,2020)	~Cumulative mortality rate (CMR), ~spatial autocorrelation,~ hot spot analysis (Getis-Ord Gi* statistic), Anselin Local Moran's	Studying the spatial distribution of COVID-19 cases globally at the end of March and April 2020	Identification of high and low-risk clusters of COVID-19 cases, determining high-risk districts	By the end of March, high CIR countries included San Marino, Ghana, Andorra, and Faroe Island. By the end of April, Ghana, San Marino, Andorra, and Luxembourg had high CIR.

Table 6:Summary of Literature Reviews presented for Hotspot Mapping

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